



Hyrule Math Quest

Grade 5 Pre-Test • Roman Numerals, Decimals & Properties



Hero's Name: Date:

- 1 The cornerstone of the Temple of Time is carved with **MCMLXXXVI**. Write that year in standard form.

Standard form:

- 2 Treasure chest inventory. Fill in every empty box.

Standard Form	Roman Numerals
44	
	XCV
360	
	MDCCXII

- 3 The royal census counts **8,364,572** blades of grass in Hyrule Field.

- a. What is the value of the digit 3?
- b. Which digit is in the ten-thousands place?
- c. Which digit is in the millions place?

- 4 The Master Sword's energy meter reads **9.628**.

- a. Which digit is in the thousandths place?
- b. What place is the 6 in?
- c. What is the value of the digit 2?

- 5 The Sheikah Slate clocks Link's shrine run at **8.476** minutes.

- a. Round it to the nearest tenth:
- b. Round it to the nearest hundredth:
- c. Now round **2.97** to the nearest tenth:



6 Reading the ancient runes.

- a. Write five and forty-seven thousandths in standard form:
- b. Write **0.315** in words:
-

7 Three potions are weighed on the shop scale. Write **<**, **>**, or **=** in each box.

0.7 **0.68**

4.320 **4.32**

2.109 **2.19**

8 Two rupee hoards sit in the Royal Vault. Circle the larger hoard.

5,408,297

and

5,480,279

I knew because the digits are different in the place.

9 Four ancient runes are charging at these levels:

3.4, **3.045**, **3.405**, **3.05**. Order them from least to greatest.

< < <

10 The Sheikah Slate has translated four ancient equations. Name the property each one shows.

Word bank:

Commutative

Associative

Distributive

Identity

- a. **$(7 \times 25) \times 4 = 7 \times (25 \times 4)$**
- b. **$6 \times 43 = 43 \times 6$**
- c. **$8 \times (50 + 3) = (8 \times 50) + (8 \times 3)$**
- d. **$348 + 0 = 348$**



11

Shortcuts through Hyrule Field.

- a. Link picks up four rupees worth $25 + 78 + 75 + 22$. **Reorder and regroup them to make the adding easier**, then find the total.

Show your regrouping

Total:

- b. Each of Link's 7 quivers holds 104 arrows. **Break 104 apart to finish the shortcut.**

$$7 \times 104 = 7 \times (\quad + \quad) = \quad + \quad = \quad$$



12. The Final Quest • Brewing the Great Elixir

Link is brewing an elixir at the cooking pot. He drops in all five ingredients below. **What is the total weight of the elixir?** Show your work.

Hearty Truffle

2.375 lb

Swift Carrot

0.84 lb

Silent Shroom

1.006 lb

Hydromelon

3.25 lb

Sunshroom

0.019 lb

Show your work here

Total weight of the elixir:

◆ Quest complete! You are a true Hero of Hyrule. ◆



1. MCMLXXXVI = **1,986** ($M = 1000$, $CM = 900$, $LXXX = 80$, $VI = 6$)
2. Treasure chest inventory:
 - **44 = XLIV** ($40 + 4$)
 - **XCV = 95** ($90 + 5$)
 - **360 = CCCLX** ($300 + 50 + 10$)
 - **MDCCXII = 1,712** ($1000 + 500 + 200 + 10 + 2$)
3. 8,364,572
 - a. Value of the 3: **300,000** (*it sits in the hundred-thousands place*)
 - b. Ten-thousands place: **6** c. Millions place: **8**
4. 9.628
 - a. Thousandths place: **8** b. The 6 is in the **tenths** place
 - c. Value of the 2: **0.02** (*two hundredths*)
5. Rounding:
 - a. 8.476 to the nearest tenth = **8.5** (*hundredths digit is 7, so the tenths place rounds up*)
 - b. 8.476 to the nearest hundredth = **8.48** (*thousandths digit is 6, so the hundredths place rounds up*)
 - c. 2.97 to the nearest tenth = **3.0** (*9 tenths rounds up and carries into the ones place. Accept 3, though 3.0 shows the tenths place. Watch for 2.10 – that means they rounded the 9 in place instead of carrying.*)
6. Reading the runes:
 - a. **5.047** (*forty-seven thousandths = 0.047, so the tenths place is a zero*)
 - b. **three hundred fifteen thousandths**
7. Comparing potions:
 - **0.7 > 0.68** (0.700 vs 0.680)
 - **4.320 = 4.32** (*the trailing zero adds no value*)
 - **2.109 < 2.19** (2.109 vs 2.190)
8. The larger hoard is **5,480,279**. Both start with 5,4, but the **ten-thousands** place decides: **8 > 0**.
9. Least to greatest: **3.045 < 3.05 < 3.4 < 3.405** *Rewrite to thousandths: 3.045, 3.050, 3.400, 3.405.*
10. Name the property:
 - a. **Associative** (*same order, regrouped parentheses*)
 - b. **Commutative** (*factors swapped order*)
 - c. **Distributive** (*the 8 is handed to both parts of $50 + 3$*)
 - d. **Identity** (*adding 0 leaves the number unchanged*)
11. Shortcuts:
 - a. **$(25 + 75) + (78 + 22) = 100 + 100 = 200$** *Accept any correct regrouping that reaches 200.*
 - b. **$7 \times (100 + 4) = 700 + 28 = 728$**
12. The Great Elixir: **7.490 lb** (or 7.49 lb)
 $2.375 + 0.840 + 1.006 + 3.250 + 0.019 = 7.490$
Adding zeros so every number reaches the thousandths place keeps the columns lined up.